KOKI HATTORI

Curriculum Vitae

Ph.D. Candidate, Department of Electrical Engineering and Information Systems
Graduate School of Engineering, The University of Tokyo, Tokyo, Japan

k.hattori@ctl.t.u-tokyo.ac.jp | LinkedIn | ORCID | Updated: 10 Nov. 2026

RESEARCH PROFILE

Doctoral researcher specializing in mechatronics, precision control, and pneumatic systems. My research focuses on achieving high-performance control of pneumatic and mechatronic systems by integrating physics-based modeling, experimental system identification, nonlinear control, and data-driven learning control.

RESEARCH INTERESTS

- High-precision and high-speed control of mechatronic systems
- Modeling and control of pneumatic systems and actuators
- Nonlinear and position-dependent dynamics modeling
- System identification and model-based feedforward control
- Nonlinear feedback control based on input-output linearization
- Data-driven control based on iterative learning control

EDUCATION

Apr. 2026–Present **Doctoral Program in Electrical Engineering and Information Systems**
Graduate School of Engineering, The University of Tokyo, Tokyo, Japan
Expected completion: March 2029.

Apr. 2024–Mar. 2026 **Master of Engineering in Electrical Engineering and Information Systems**
Graduate School of Engineering, The University of Tokyo, Tokyo, Japan
Thesis: Precise Flow Rate Control for High-Speed Operation of Pneumatic Systems.

Apr. 2020–Mar. 2024 **Bachelor of Engineering in Electrical and Electronic Engineering**
Faculty of Engineering, The University of Tokyo, Tokyo, Japan
Thesis: Precise Flow Rate Control Based on Effective Orifice Area of Pneumatic Valves.

RESEARCH EXPERIENCE

Apr. 2024–Present **Graduate Researcher, Koseki–Ohnishi Laboratory**
The University of Tokyo, Tokyo, Japan
Research on high-precision and high-speed control of pneumatic systems through model-based nonlinear compensation and data-driven learning control.

PUBLICATIONS

Peer-Reviewed Journal Articles

[J1] **K. Hattori**, W. Ohnishi, and T. Koseki, "Feedforward Flow Rate Control for Pipe Resonance Cancellation via Iterative Learning-Based Parameter Identification," *IEEJ Journal of Industry Applications*, vol. 15, no. 3, pp. 375–384, 2026. doi:10.1541/ieejjia.20250331.

[J2] **K. Hattori**, W. Ohnishi, and T. Koseki, "High-precision and high-speed flow rate control with nonlinear disturbance compensation using valve body position feedback," *Mechatronics*, vol. 108, p. 103317, 2025. doi:10.1016/j.mechatronics.2025.103317.

Peer-Reviewed International Conference Papers

[C1] **K. Hattori**, W. Ohnishi, and T. Koseki, "Pressure Fluctuation Suppression and Precise Flow Rate Control through Simultaneous Control for Series-connected Two-port Valves," in *Proceedings of the 23rd IFAC World Congress*, 2026.

[C2] K. Goto, **K. Hattori**, K. Hashimoto, I. Kishimoto, T. Koseki, and W. Ohnishi, "Model Based Feedforward Control for Motion Stage Reaction Force Cancellation of Pneumatic Vibration Isolation Tables," in *2026 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM2026)*, pp. 1–6, 2026.

[C3] K. Kuroda, **K. Hattori**, W. Ohnishi, and T. Koseki, "Modeling of a Long Tube for Mass Flow Rate Control and Its Parameter Estimation via Iterative Learning Control," in *12th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON2026)*, pp. 96–101, 2026.

[C4] K. Goto, **K. Hattori**, K. Hashimoto, I. Kishimoto, T. Koseki, and W. Ohnishi, "Quick Response Valve for 3DOF Active Control of Pneumatic Vibration Isolation Tables," in *51st Annual Conference of the IEEE Industrial Electronics Society (IECON 2025)*, pp. 1–6, 2025.

[C5] E. Shoshi, **K. Hattori**, T. Ibrahim, F. P. Dambo, T. Koseki, and W. Ohnishi, "Flow Rate Ripple Suppression Based on Phase Shift Optimization of Two PWM-Actuated High-Speed Valves," in *2025 Joint IFAC Symposium on Mechatronic Systems & Robotics*, pp. 233–238, 2025.

[C6] **K. Hattori**, W. Ohnishi, and T. Koseki, "Modeling of Airflow Disturbance Force and Feedforward Controller Design in Pneumatic Valves," in *2025 IEEE International Conference on Mechatronics (ICM2025)*, pp. 1–6, 2025.

[C7] K. Goto, **K. Hattori**, K. Hashimoto, I. Kishimoto, T. Koseki, and W. Ohnishi, "Quick Response Valve for Active Control of Pneumatic Vibration Isolation Tables," in *2025 IEEE International Conference on Mechatronics (ICM2025)*, pp. 1–6, 2025.

[C8] **K. Hattori**, W. Ohnishi, and T. Koseki, "Compensation of Pressure Dependent Disturbance: Poppet Position Control in a Pneumatic Valve," in *2024 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM2024)*, pp. 1–6, 2024.

[C9] **K. Hattori**, W. Ohnishi, and T. Koseki, "Mass Flow Rate Control with Compensation of Nonlinearity Using Valve Internal Variables," in *2024 IEEE 18th International Conference on Advanced Motion Control (AMC2024)*, pp. 1–6, 2024.

AWARDS AND HONORS

19 Jun. 2026	Hiroshi Harashima Academic Encouragement Award, Electrical and Electronic Information Science Promotion Foundation.
24 Mar. 2026	Dean's Award, Graduate School of Engineering, The University of Tokyo.
24 Mar. 2026	Outstanding Master's Thesis Award, Department of Electrical Engineering and Information Systems, Graduate School of Engineering, The University of Tokyo.
30 Jan. 2026	Encouragement Award, Technical Committee on Linear Drives, The Institute of Electrical Engineers of Japan (IEEJ), for a presentation at the Joint Technical Meeting on Linear Drives and Transportation and Electric Railway.
1 Mar. 2024	Best Paper Award, 2024 IEEE 18th International Conference on Advanced Motion Control (AMC2024).

FELLOWSHIPS, SCHOLARSHIPS, AND RESEARCH GRANTS

Apr. 2026–Mar. 2029	Selected Project Student, *Fostering Advanced Human Resources to Lead Green Transformation (GX) (SPRING GX)*, The University of Tokyo / Japan Science and Technology Agency.
Jul. 2025	International Technical Exchange Assistance (Overseas Travel and Stay), TEPCO Memorial Foundation.
Oct. 2024–Mar. 2026	Selected Project Student, *World-leading Innovative Graduate Study Program for Quantum and Semiconductor Science & Technology Program (WINGS-QSTEP)*, The University of Tokyo.
Jul. 2024	Overseas Research Grant, The Murata Science and Education Foundation.
Jul. 2024	Student Travel Award, IEEE Industrial Electronics Society.

RESEARCH VISITS

Oct. 2026–Dec. 2026 **Visiting Scholar, Institute for System Dynamics**
University of Stuttgart, Stuttgart, Germany
Research on a pneumatically driven flight simulator motion platform.

TECHNICAL SKILLS

Control Engineering	Feedback control, loop shaping, model-based feedforward control, iterative learning control, frequency response analysis, experimental system identification
Pneumatics	Pneumatic system modeling, pneumatic actuator and valve modeling, flow rate and pressure control, experimental system design
Mechatronics	Real-time control implementation, actuator and sensor integration, hardware-software integration
Programming & Software	MATLAB, Simulink, C/C++, Beckhoff TwinCAT 3, dSPACE ControlDesk
Languages	Japanese (native); English (professional working proficiency)